AP Physics 1 - Unit 3 - Lesson 1

What is energy? Types of energy

WORKED SOLUTIONS + NOTES

Teacher/student reference. These notes mirror the HTML walkthrough and the student work-pack. They intentionally introduce energy quantities before the mechanical-work equation.

Unit overview

L1	What is energy? Types of energy
L2	Energy transfers and conservation of energy
L3	Power of an energy-transfer machine
L4	Mechanical work
L5	Spring-energy derivation
L6	Work-energy theorem and systems
L7	Applications: inclines, Atwood systems, etc.
L8	Assessed lab
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L10	UNIT TEST - 16/17 November 2026

Proposed test: 16/17 November 2026

Full notes 1 - What is energy?

Energy is a scalar quantity used to account for the state of an object or system and the changes that can occur. Energy can be transferred between systems and transformed between different stores/forms. In every interaction, total energy is conserved.

SI unit: joule (J). One joule is equivalent to 1 kg m^2 s^{-2}.

GREENSICK memory aid

Letter	Form/store	Useful classification
G	Gravitational	potential
R	Rotational	kinetic
E	Elastic	potential
E	Electrical	context-dependent / interaction energy
N	Nuclear	interaction/internal
S	Sound	wave-carried energy
I	Internal	microscopic kinetic + potential
C	Chemical	microscopic potential/internal
K	Kinetic	kinetic

Important conceptual distinction: “kinetic” and “potential” are broad physics categories. Translational and rotational energy are kinetic because they depend on motion. Gravitational and elastic/spring energies are potential because they belong to interacting systems/configurations. Many familiar labels such as chemical, nuclear and internal energy are deeper combinations of microscopic kinetic and potential energies.

Full notes 2 - AP-style energy symbols and equations

Translational kinetic energy

$$K = \tfrac{1}{2}mv^2$$

SI quantities: m in kg; v in m s^{-1}

Current Unit 3 AP equation.

Rotational kinetic energy

$$K_R = \tfrac{1}{2}I\omega^2$$

SI quantities: I in kg m^2; ω in rad s^{-1}

AP equation developed formally in Unit 6; introduced here as part of the energy family.

Near-surface gravitational PE change

$$\Delta U_g = mg\Delta y$$

SI quantities: m in kg; g in N kg^{-1}; Δy in m

Use when g is approximately constant.

General gravitational PE

$$U_g = -Gm_1m_2/r$$

SI quantities: G = 6.67 × 10^{-11} N m^2 kg^{-2}; r in m

Zero is defined at infinite separation.

Spring potential energy

$$U_s = \tfrac{1}{2}k(\Delta x)^2$$

SI quantities: k in N m^{-1}; Δx in m

Elastic energy stored in an ideal spring.

Internal-energy change

$$\Delta U_{int} = U_{int,f} - U_{int,i}$$

SI quantities: all energy values in J

Useful bookkeeping notation. The current AP Physics 1 CED requires awareness of dissipation to thermal/sound but does not print a separate ΔU_int equation.

Mechanical energy

$$E_{mech} = K + U$$

SI quantities: all terms in J

The sum of kinetic and potential energies of the chosen system.

Potential energy belongs to a system. For example, gravitational potential energy is associated with the object-Earth system, not with the object by itself. The zero of potential energy may be chosen for convenience in near-surface problems; for U_g = $-Gm_1m_2/r$, the conventional zero is at infinite separation.

Mechanical energy: $E_{mech} = K + U$. If the system has more than one kinetic or potential contribution, add them all. For a rolling object later in the course, for example, total K can contain both translational and rotational kinetic energy.

Round A - direct substitution in SI

A1. Translational kinetic energy
A 2.0 kg cart moves at 3.0 m s^{-1}. Determine its translational kinetic energy.

Useful hint: Use $K = \frac{1}{2}mv^2$. All quantities are already in SI units.

1. $K = \frac{1}{2}mv^2$
2. $K = \frac{1}{2}(2.0\ kg)(3.0\ m\ s^{-1})^2$
3. $K = 9.0\ J$

Answer: 9.0 J

A2. Rotational kinetic energy
A wheel has rotational inertia $I = 0.50\ kg\ m^2$ and angular speed $\omega = 4.0\ rad\ s^{-1}$. Determine its rotational kinetic energy.

Useful hint: Use $K_R = \frac{1}{2}I\omega^2$.

1. $K_R = \frac{1}{2}I\omega^2$
2. $K_R = \frac{1}{2}(0.50)(4.0)^2$
3. $K_R = 4.0\ J$

Answer: 4.0 J

A3. Gravitational PE change
A 3.0 kg object is raised vertically by 2.0 m near Earth. Use $g = 9.8\ N\ kg^{-1}$. Determine ΔU_g.

Useful hint: Use $\Delta U_g = mg\Delta y$. Raising the object means Δy is positive.

1. $\Delta U_g = mg\Delta y$
2. $\Delta U_g = (3.0)(9.8)(+2.0)$
3. $\Delta U_g = +58.8\ J$

Answer: +58.8 J

A4. General gravitational PE
A 1000 kg satellite is 7.00×10^6 m from Earth's center. Take Earth's mass as 5.97×10^{24} kg. Determine U_g.

Useful hint: Use $U_g = -Gm_1m_2/r$. Gravitational potential energy is negative with zero at infinite separation.

1. $U_g = -Gm_1m_2/r$
2. $U_g = -(6.67 \times 10^{-11})(5.97 \times 10^{24})(1000)/(7.00 \times 10^6)$
3. $U_g \approx -5.69 \times 10^{10}\ J$

Answer: -5.69×10^{10} J

A5. Spring potential energy
An ideal spring has $k = 200\ N\ m^{-1}$ and is compressed by 0.10 m. Determine U_s.

Useful hint: Use $U_s = \frac{1}{2}k(\Delta x)^2$. The sign of Δx does not matter because it is squared.

1. $U_s = \frac{1}{2}k(\Delta x)^2$
2. $U_s = \frac{1}{2}(200)(0.10)^2$
3. $U_s = 1.00\ J$

Answer: 1.00 J

A6. Internal-energy change

A system’s internal energy changes from 250 J to 310 J. Determine ΔU_{int}.

Useful hint: Change = final − initial.

1. $\Delta U_{int} = U_{int,f} - U_{int,i}$
2. $\Delta U_{int} = 310\ J - 250\ J$
3. $\Delta U_{int} = +60\ J$

Answer: +60 J

A7. Mechanical energy

A system has K = 20 J, U_g = 35 J, and U_s = 5 J. Determine E_{mech}.

Useful hint: Mechanical energy is the sum of the kinetic and potential energies in the selected system.

1. $E_{mech} = K + U$
2. $E_{mech} = 20 + 35 + 5$
3. $E_{mech} = 60\ J$

Answer: 60 J

Round B - convert to SI first

B1. Translational kinetic energy
A 750 g ball moves at 18 km h^{-1}. Determine K in joules.

Useful hint: Convert 750 g to kg and 18 km h^{-1} to m s^{-1} before using $K = \frac{1}{2}mv^2$.

1. m = 750 g = 0.750 kg

2. v = 18 km h^{-1} ÷ 3.6 = 5.0 m s^{-1}

3. $K = \frac{1}{2}(0.750)(5.0)^2 = 9.375\ J \approx 9.4\ J$

Answer: 9.4 J

B2. Rotational kinetic energy
A rotor has I = 0.020 kg m^2 and rotates at 120 rpm. Determine K_R.

Useful hint: Convert rpm to rad s^{-1}: 120 rpm = 2 rev s^{-1} = 4π rad s^{-1}.

1. $\omega = 120\ rev\ min^{-1} \times (1\ min/60\ s) \times 2\pi\ rad/rev = 4\pi\ rad\ s^{-1}$

2. $K_R = \frac{1}{2}I\omega^2$

3. $K_R = \frac{1}{2}(0.020)(4\pi)^2 \approx 1.58\ J$

Answer: 1.58 J

B3. Gravitational PE change
A 650 g mass is lifted through 45 cm. Use g = 9.8 N kg^{-1}. Determine ΔU_g.

Useful hint: Convert grams to kilograms and centimetres to metres.

1. m = 0.650 kg; Δy = 0.45 m

2. $\Delta U_g = mg\Delta y$

3. $\Delta U_g = (0.650)(9.8)(0.45) \approx 2.87\ J$

Answer: 2.87 J

B4. General gravitational PE
A 2.0 tonne satellite is 7000 km from Earth's center. Use Earth mass 5.97×10^{24} kg. Determine U_g.

Useful hint: 2.0 tonnes = 2000 kg and 7000 km = 7.00×10^6 m.

1. $m_2 = 2000\ kg$; $r = 7.00 \times 10^6\ m$

2. $U_g = -Gm_1m_2/r$

3. $U_g \approx -1.14 \times 10^{11}\ J$

Answer: -1.14×10^{11} J

B5. Spring potential energy
A spring has k = 3.0 N cm^{-1} and is compressed by 6.0 cm. Determine U_s.

Useful hint: Convert k: 1 N cm^{-1} = 100 N m^{-1}. Convert 6.0 cm to 0.060 m.

1. k = 3.0 N cm^{-1} = 300 N m^{-1}; Δx = 0.060 m

2. $U_s = \frac{1}{2}k(\Delta x)^2$

3. $U_s = \frac{1}{2}(300)(0.060)^2 = 0.54\ J$

Answer: 0.54 J

B6. Internal-energy change

A system’s internal energy changes from 0.45 kJ to 620 J. Determine ΔU_{int} in joules.

Useful hint: Convert 0.45 kJ to joules first.

1. $U_{int,i} = 0.45\ \text{kJ} = 450\ \text{J}$
2. $\Delta U_{int} = 620 - 450$
3. $\Delta U_{int} = +170\ \text{J}$

Answer: +170 J

B7. Mechanical energy

A system has K = 0.80 kJ, U_g = 350 J, and U_s = 0.050 kJ. Determine E_{mech} in joules.

Useful hint: Convert both kJ values to J, then add all kinetic and potential terms.

1. $K = 800\ \text{J}$; $U_s = 50\ \text{J}$
2. $E_{mech} = 800 + 350 + 50$
3. $E_{mech} = 1200\ \text{J}$

Answer: 1200 J

Round C - rearrange first

C1. Translational kinetic energy
An 8.0 kg object has K = 64 J. Determine its speed.

Useful hint: Rearrange $K = \frac{1}{2}mv^2$ to $v = \sqrt{(2K/m)}$.

1. $v = \sqrt{(2K/m)}$

2. $v = \sqrt{[(2)(64)/8.0]}$

3. $v = 4.0\ \text{m s}^{-1}$

Answer: $4.0\ \text{m s}^{-1}$

C2. Rotational kinetic energy
A rotor has $K_R = 9.0$ J and $I = 0.50\ \text{kg m}^2$. Determine ω.

Useful hint: Rearrange $K_R = \frac{1}{2}I\omega^2$ to $\omega = \sqrt{(2K_R/I)}$.

1. $\omega = \sqrt{(2K_R/I)}$

2. $\omega = \sqrt{[(2)(9.0)/0.50]}$

3. $\omega = 6.0\ \text{rad s}^{-1}$

Answer: $6.0\ \text{rad s}^{-1}$

C3. Gravitational PE change
A 5.0 kg object gains 196 J of gravitational potential energy near Earth. Use $g = 9.8\ \text{N kg}^{-1}$. Determine Δy.

Useful hint: Rearrange $\Delta U_g = mg\Delta y$.

1. $\Delta y = \Delta U_g/(mg)$

2. $\Delta y = 196/[(5.0)(9.8)]$

3. $\Delta y = 4.0$ m

Answer: 4.0 m

C4. General gravitational PE
A 1.00×10^3 kg spacecraft has $U_g = -6.67 \times 10^9$ J in the field of a planet of mass 5.00×10^{24} kg. Determine r.

Useful hint: Use the magnitude of U_g when rearranging: $r = Gm_1m_2/|U_g|$.

1. $r = Gm_1m_2/|U_g|$

2. $r = (6.67 \times 10^{-11})(5.00 \times 10^{24})(1.00 \times 10^3)/(6.67 \times 10^9)$

3. $r = 5.00 \times 10^7$ m

Answer: 5.00×10^7 m

C5. Spring potential energy
A spring with $k = 100\ \text{N m}^{-1}$ stores 4.5 J. Determine $|\Delta x|$.

Useful hint: Rearrange $U_s = \frac{1}{2}k(\Delta x)^2$ to $|\Delta x| = \sqrt{(2U_s/k)}$.

1. $|\Delta x| = \sqrt{(2U_s/k)}$

2. $|\Delta x| = \sqrt{[(2)(4.5)/100]}$

3. $|\Delta x| = 0.30$ m

Answer: 0.30 m

C6. Internal-energy change
A system starts with $U_{int,i} = 120$ J and has $\Delta U_{int} = +45$ J. Determine $U_{int,f}$.

Useful hint: From $\Delta U_{int} = U_{int,f} - U_{int,i}$, rearrange to $U_{int,f} = U_{int,i} + \Delta U_{int}$.

1. $U_{int,f} = U_{int,i} + \Delta U_{int}$
2. $U_{int,f} = 120 + 45$
3. $U_{int,f} = 165$ J

Answer: 165 J

C7. Mechanical energy
A system has $E_{mech} = 90$ J, $K = 35$ J, and $U_g = 40$ J. The only other mechanical-energy store is a spring. Determine U_s.

Useful hint: Use $E_{mech} = K + U_g + U_s$ and rearrange.

1. $U_s = E_{mech} - K - U_g$
2. $U_s = 90 - 35 - 40$
3. $U_s = 15$ J

Answer: 15 J

Common issues to watch for

• Potential energy is a system quantity. Avoid saying “the object has gravitational potential energy” when you are being precise; it is the object-Earth (or two-mass) system.

• ΔU_g can be positive or negative. Its sign follows Δy in a constant gravitational field.

• General U_g is negative. With zero at infinite separation, a bound two-mass gravitational system has $U_g < 0$.

• Spring energy is never negative for the ideal expression $\frac{1}{2}k(\Delta x)^2$, because displacement from equilibrium is squared.

• Convert before substituting. Grams, centimetres, kilometres, tonnes, km h^{-1} and rpm must be converted to SI.

• Mechanical energy is not “all energy”. It is specifically the sum of kinetic and potential energies of the chosen system.

• Rotational kinetic energy is previewed here. Its full treatment belongs to Unit 6.

• ΔU_{int} notation: useful for bookkeeping when energy becomes microscopic/internal. The current AP Physics 1 CED does not print a standalone ΔU_{int} equation.

Suggested exit question

A ball is moving upward. Without doing any calculation, identify which AP energy quantities could be changing and state what extra information you would need before assigning numerical values.